Sailing Around the Carbon Tax: Vessel Behaviour Under the EU Emissions Trading System

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2. Literature Review
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Introduction

- International shipping accounts for roughly 2% of global CO_2 emissions , with bulk shipping accounting for around 20% of these emissions
- Textbook negative externality
- The maritime sector entered the EU Emissions Trading System (EU ETS) in January 2024.
- ETS coverage is asymmetric:
  - 100% of emissions on intra-EU voyages
  - 50% of emissions on voyages between EU and non-EU ports

How did the 2024 extension of the EU ETS to the shipping fleet affect bulk vessel behaviour?

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Literature Review

Partial regulation → rerouting

- Ships comply inside the regulated zone but reroute outside it to avoid costs
  - Klotz & Berazneva (2022), Hansen-Lewis & Marcus (2025): only 53% of expected cuts materialised under sulphur rules
  - We apply this logic to carbon under the EU ETS

EU ETS leakage: predicted, never measured

- The 50/100 rule makes a non-EU port stop financially rational
  - Wu et al. (2024): modelled as a prisoner's dilemma
  - Our paper want's to test if this really shows in real post-2024 data

Literature Review

Pre-reform elasticity estimates

- How would shipping respond to a carbon price?
  - Mundaca & Strand (2020): Estimated before the EU ETS existed: adjustment through less trade, not emission reduction
  - We put real post-2024 data behind these estimates

Vessel behaviour under price signals

- Contract structure mediates responses more than the price itself
  - Adland & Jia (2018): the charterer pays fuel and decides speed, limiting observable compliance behaviour
  - Economides (2025): vessel-level DiD finds 17% speed drop under a port policy
  - We extended this design towards a continuous treatment

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Institutional Background: What Matters for Our Data

The institutional setting matters because the policy creates **different exposure intensities** across vessels.

Key empirical implication

Voyages are not simply treated or untreated:

- intra-EU voyages receive full ETS exposure;
- EU $\leftrightarrow$ non-EU voyages receive partial exposure;
- non-EU voyages receive no exposure.

- We translate this rule into a continuous monthly variable: **ETS_Rate**.
- Because the policy was announced before 2024, some pre-reform anticipation is possible.
- This is why we later check **2023 pre-trends** before estimating the policy effect.

Data and Descriptive Statistics

Raw data: voyage-level commercial shipping records, January 2013–June 2025.

- Around **3.8 million voyages**, aggregated to a monthly vessel panel.
- Final balanced sample: **15,781 vessels** and **1,755,789 vessel-month observations**.

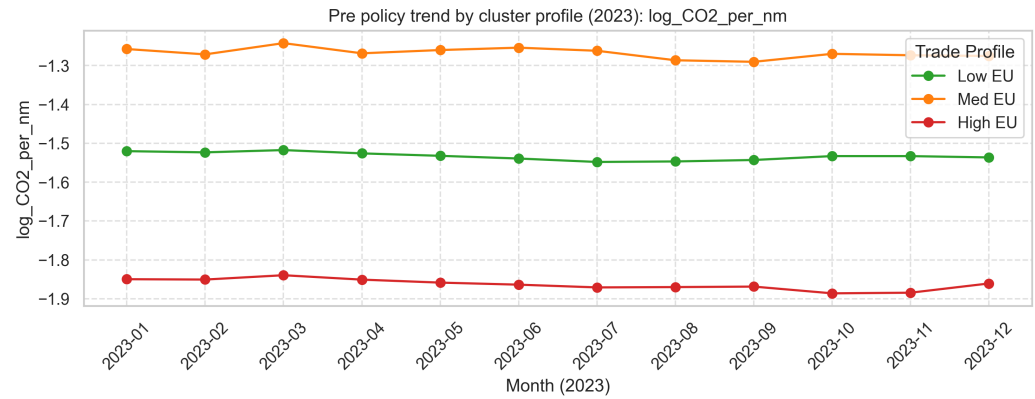
Variable	Mean	Median	P25	P75
log(CO_2/nm)	-1.561	**-1.542**	-1.966	-1.233
Avg Speed (kn)	10.97	**11.15**	9.89	12.18
Distance (nm)	4,136	**3,093**	1,319	5,478
MGO Share	0.033	0.016	0.008	0.038
HSFO Share	0.064	**0.000**	0.000	0.000
VLSFO Share	**0.903**	**0.981**	0.954	0.992

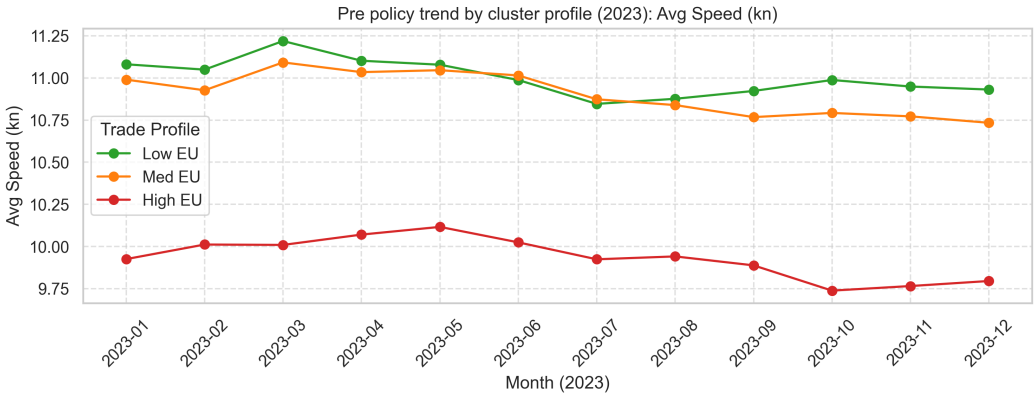
- Median vessel-month: **3,093 nm**, **11.15 knots**, around **0.21 tonnes CO_2/nm**.
- Fuel mix is dominated by **VLSFO**; HSFO use is concentrated among scrubber-equipped vessels.

Trade Profiles and Pre-Reform Trends

Variable	High	Med	Low
Vessels	13,664	11,465	15,056
log CO_2/nm	-1.860	-1.234	-1.516
Speed	**10.18**	11.32	11.32
Distance	**2,805**	**5,697**	4,274
MGO	**0.047**	0.023	0.029
HSFO	0.025	**0.124**	0.062
VLSFO	**0.928**	0.853	0.909

Interpretation: High EU vessels sail shorter routes and use more MGO; Med EU vessels operate the longest routes and have the highest HSFO share.





2023 pre-reform trends are broadly stable before January 2024.

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The Identification Challenge

Problem: The EU ETS hit every vessel at the same date

- No ship was untreated $\Rightarrow$ no clean control group $\Rightarrow$ standard DiD fails

Solution: Exploit variation in exposure intensity instead

$\text{ETS_Rate}_{it} \in [0, 1]$: how much of a vessel's trade is actually taxed

- Intra-EU voyages: **1.0** (full coverage)
- EU $\leftrightarrow$ non-EU voyages: **0.5** (partial coverage)
- Non-EU voyages: **0.0** (no coverage)

The DiD treatment variable combines exposure with the reform date:

$$\textit{Interaction}_{it} = \underbrace{\textit{ETS_Rate}_{it}}_{\text{who is exposed}} \times \underbrace{\textit{Post2024}_{t}}_{\text{when the policy started}}$$

Two Specifications

Both models use Two Way Fixed Effects (vessel FE + month FE)

Model A - baseline DiD:

$$Y_{it} = \beta_A \cdot Interaction_{it} + \delta \cdot ETS_Rate_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Model B - main specification (adds a trend control):

$$Y_{it} = \beta_B \cdot Interaction_{it} + \delta \cdot ETS_Rate_{it} + \theta \cdot (ETS_Rate_{it} \times TimeIdx_t) + \alpha_i + \gamma_t + \varepsilon_{it}$$

Fixed effects

α_i: vessel size, type, trade route

γ_t: Red Sea, freight cycles, seasonality

SE clustered at the vessel level

β_B: what we care about

The break at Jan 2024, after filtering out any pre-existing slope proportional to EU exposure

Fleet Segmentation: Three Trade Profiles

Problem: Global averages can hide opposite effects across vessel types
Solution: PCA + Ward hierarchical clustering on 2023 country level data

1. Country level means of 7 features: distance, CO_2 intensity, EU exposure, speed, DWT, MGO share, laden ratio
2. PCA $\rightarrow$ Ward clustering $\rightarrow$ 3 groups, relabeled by EU exposure
3. Each vessel inherits the cluster of its departure country that month

Low EU	**Med EU**	**High EU**
Near-zero ETS cost	Deep-sea bulk fleet	Mostly intra-EU
Moderate routes	Longest routes (5,700 nm)	Short routes (2,800 nm)
15,056 vessels	Large vessels (90k DWT)	Highest ETS bill

Note: Med EU is not "medium exposure", but it is the long-haul fleet that occasionally calls at EU ports. What separates it from Low EU is vessel size and route length, not ETS exposure.

Threats to Identification

- **Parallel trends:**
  Pre-trend plots over 2023 show all three clusters tracking each other closely
- **Red Sea crisis (late 2023):**
  Month FE absorbs the common shock. Model B's trend control handles any differential effect.
- **Anticipation effects:**
  30 month gap between announcement (July 2021) and start (Jan 2024), which is why no RDD where used.
  $\Rightarrow \hat{\beta}$ is a *lower bound* on the true effect.
- **IMO CII regulation (Jan 2023):**
  Applies to all vessels equally $\Rightarrow$ fully absorbed by month FE. **Not a threat.**

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Why Specification Choice Matters — Global Results

	Model A (baseline)		Model B (main spec)	
Outcome variable	Coef.	p-value	Coef.	p-value
$\log(CO_2/nm)$	0.0167	0.000***	0.0039	0.020**
MGO share	0.0005	0.010***	−0.0001	0.667
HSFO share	−0.0002	0.106	0.0001	0.109
VLSFO share	−0.0004	0.126	0.0000	0.939
Avg speed (kn)	0.1796	0.000***	0.0102	0.448
Distance (nm)	286.55	0.000***	57.57	0.048**

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Cluster-Level Results – The Full Picture

	Low EU		Med EU		High EU	
Outcome variable	Coef.	p-value	Coef.	p-value	Coef.	p-value
$\log(CO_2/nm)$	−0.0227	0.015**	−0.0681	0.000***	0.0404	0.000***
MGO share	−0.0005	0.704	−0.0003	0.624	0.0037	0.000***
HSFO share	0.0001	0.824	−0.0012	0.034**	0.0000	0.200
VLSFO share	0.0005	0.734	0.0015	0.035**	−0.0037	0.000***
Avg speed (kn)	−0.1576	0.001***	−0.3039	0.000***	−0.1141	0.000***
Distance (nm)	772.96	0.000***	362.94	0.052*	−413.04	0.000***

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

- **Finding 1:** speed reduction across all three clusters
- **Finding 2:** route restructuring & leakage
- **Finding 3:** fuel switching
- **Finding 4:** the CO_2 intensity paradox

Finding 1: Speed Reduction Across All Three Clusters

All three clusters **significantly reduced average speed** post-reform:

- **Med EU:** -0.3039 **kn** (largest) — long routes give flexibility to extend transit time without disrupting scheduling; also faces CII pressure which makes slow pacing more efficient
- **Low EU:** -0.1576 **kn** — consistent with general equilibrium congestion effects as High EU cargo diverts and lower approach speed reduces congestion costs
- **High EU:** -0.1141 **kn** — consistent with EU ETS cost incentive to slow steam

Note on identification

Coefficients identify *within-cluster* differential speed response by ETS_Rate intensity

Red Sea common shock is fully absorbed by month fixed effects

$\Rightarrow$ Results cannot be attributed to Red Sea rerouting

Finding 2: Route Restructuring & Leakage

Asymmetric distance responses:

- **High EU:** -413.04 **nm** (-15%)
  Fewer taxed nautical miles $\Rightarrow$ fewer allowances required; achievable by adding non-EEA intermediate stop
- **Low EU:** $+772.96$ **nm** $(+21\%)$
  Expanding routes after Jan 2024 points to cargo diversion, not a global shock
- **Med EU:** $+362.94$ **nm**
  Some vessels benefit similarly to Low EU
  Marginal significance $\Rightarrow$ greater heterogeneity within this cluster

Carbon Leakage

High EU contracts, Low EU expands by nearly twice as much

Low EU vessels are more numerous $\Rightarrow$ net global nm sailed likely *increases*

Rather than lowering global shipping emissions, the EU ETS reform may well simply redistribute it

Finding 3: Fuel Switching

High EU cluster only:

- MGO share: $+0.37$ **pp** $(***)$
- VLSFO share: -0.37 **pp** $(***)$
- No significant shift in Low EU or Med EU

Mechanism: MGO has lower CO_2 content than VLSFO; High EU vessels already equipped for MGO use in EU Emission Control Areas; EU ETS deepens the cost-benefit calculus toward MGO on covered legs

Why the magnitude is small

- VLSFO dominates at 92–93% pre-reform, little room to switch
- 2024 $\Rightarrow$ effective carbon cost $\approx$ €24/tonne
- At 100% obligation (2026): effective cost $\approx$ €60/tonne, incentive 2.5$\times$ stronger

Finding 4: The CO_2 Intensity Paradox

Most heavily regulated vessels became less efficient:

- **High EU:** $+4.0\%$ CO_2/nm $(***)$ — efficiency *worsened*
- **Med EU:** -6.8% CO_2/nm $(***)$ — textbook slow steaming: longer routes, lower speed
- **Low EU:** -2.3% CO_2/nm $(**)$ — longer open-ocean routes are structurally more efficient

Explanation:
High EU's route contraction $\Rightarrow$ more frequent port calls $\Rightarrow$ emit much more CO_2 per nm than open-ocean cruising.

The Core Paradox

The EU ETS taxes *total covered emissions*

What matters for the climate is *CO_2 per unit of transport work*

These two metrics diverge: shortening routes reduces allowance cost but *raises* carbon intensity

$\Rightarrow$ The policy's compliance metric and its climate objective are **misaligned**

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Policy Implications

Fix the 50/100 rule

- Move to 100% coverage on all voyages with any EEA port call, regardless of origin
  - Eliminates the financial incentive to add a non-EU intermediate stop
  - Political obstacle: the 2012 aviation ETS was suspended within a year under UNCLOS/WTO pressure

Differentiate the obligation by trade intensity

- High EU vessels are locked into European trade and cannot reroute: lower obligation removes the incentive to try
  - Low EU vessels gain the most from diversion and should face higher pressure
  - Analogy: EU ETS Phase 3 gave free allowances to industries most at risk of relocating

Policy Implications

The policy taxes the wrong metric

- The EU ETS taxes total emissions on the European portion of their voyage, not CO_2 per unit of transport work
  - Shortening the EU portion reduces the taxed metric while worsening the climate metric
  - This misalignment persists as long as the policy tracks voyage-level emissions
  - An efficiency-linked charge would reward genuine decarbonization rather than route manipulation

The broader lesson

- For a globally mobile industry emitting a global pollutant, regional pricing alone is not enough
  - Closing the leakage channel requires either reforming the EU ETS coverage rules or advancing a coordinated global carbon levy at the IMO

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Main findings

- **Speed**: initial slowdown reflects pre-existing trends, not the carbon price
- **Routes**: clear leakage, High EU vessels shortened routes by 413 nm; Low EU expanded by 773 nm
- **Fuel**: modest shift toward cleaner MGO among the most regulated vessels
- **CO_2 efficiency**: paradoxically *worsened* for the most regulated, shorter routes mean more port stops, higher emissions per mile

Overall implication

- The policy affects behavior, but may **reallocate emissions globally rather than reduce them**.

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